

Miles Modeste

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EDUCATION

Wesleyan University, Middletown, CT May 2025

Bachelor of Arts: Computer Science, Minors: Design Engineering & Applied Sciences, East Asian Studies

Technical: Python, C, C++, C#, HTML, CSS, Java, JavaScript

Tools: GitHub, Arduino, Unity, SolidWorks, ArchiCAD, Rhino, ArcGIS

Skills: Native English and German, Intermediate Japanese, Conversational Chinese

EXPERIENCE

Research Assistant, *Wesleyan Soft Robotics Lab*, Middletown, CT Sep 2024–Sep 2025

- Prototyped emerging robotics use cases and tested alternative low-cost stretch sensors.
- Credited as co-author in forthcoming robotics & automation publication.
- Automated a data pipeline to visualize comprehensive graphs using Pandas, Matplotlib to verify sensor fiber durability under repeated use and extreme conditions.
- Reported weekly findings to a board of 11, managing tasks using GitHub in a team of 3.
- Led and mentored 2 new assistants through onboarding and integration.
- Maintained thorough, accessible technical documents for trainees and product reproducibility.

Capacitive Soft-Sensor Arm Sleeve, *Wesleyan Soft Robotics Lab* Aug 2025

- Developed a 20 soft-sensor array for integration into a flexible robotic sleeve, enabling arm behavior from human interaction and proximity sensing via I2C communication.
- Scaled sensor array from 1-20 with support up to 228 sensors using a custom data structure and improving sensing accuracy.
- Refactored C++ code, increasing capacitive range by 4x and reducing noise substantially.
- Resolved communication bottlenecks by designing binary protocol to reduce data footprint by 58% through bit packing.
- Developed Python script to collect and structure sensor input and robotic arm position data.

Using Machine Learning and Web Scraping to Understand Online Spaces, Middletown, CT May 2025

- Built a Python data pipeline using facial embeddings to quantify subject screentime, and comment sentiment analysis (Vader Sentiment) to assess online public reputation for celebrity marketing deals.
- Built a scalable web scraping system to process 1,000+ comments per video using YouTube API, performing frame-by-frame facial recognition (CNN), designed for expansion to larger datasets.
- Pandas and NumPy to structure model inputs and correlate visual presence with audience sentiment.

Obstacle Avoiding Robot, Middletown, CT May 2025

- Engineered 3x5x2in. autonomous 4 wheeled robot capable of correcting against potential crashes using a single IR proximity sensor.
- Programmed finite state machine while scanning for avoidance and recovery to manage sensor data and motor PWM signals

Studio Intern, *AJK Architekten BDA*, Frankfurt, Germany Jul 2022–Aug 2022

- Communicated in a German-speaking team to conduct an on-site measurement survey for a residential renovation project delivering accurate data necessary to finalize initial 3D-renderings.
- Created digital copies of historical floorplans and 3D-renderings for preservation in ArchiCAD.
- Applied studio-standard modeling techniques to construct high quality replica models.